

Lecture Notes Template

Problem-first structure for concise technical lectures

A reusable visual and editorial system for turning current handwritten notes, supporting transcript context, and verified course artifacts into focused technical lecture notes.

Learning goals. By the end of this lecture, you should be able to

1. formulate the practical problem and the evidence needed to evaluate it;
2. explain the method and the assumption that makes it plausible; and
3. diagnose a failure at the correct layer of the system.

1 Begin with the problem

Open with the decision or failure that motivates the lecture. Introduce the algorithm only after the problem has a clear formulation, and keep the opening example proportional to its role in the lecture.

DESIGN NOTE

Use the margin for a brief design implication, not a second narrative thread.

1.1 Move from the problem to a formulation

Show the conceptual translation explicitly:

practical problem $\rightarrow$ mathematical condition
 $\rightarrow$ model or objective $\rightarrow$ evaluation evidence.

Define notation immediately before it is used and explain why each component matters. One focused worked example is usually more valuable than several near-duplicates.

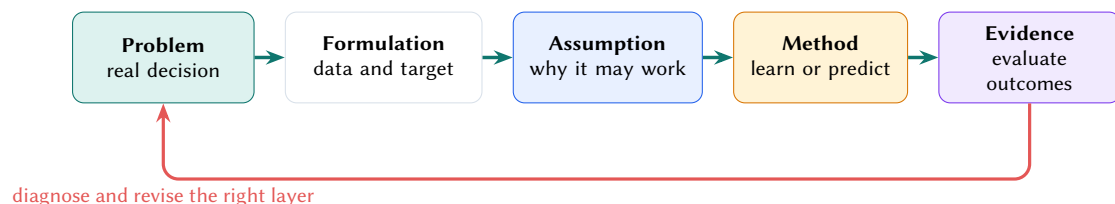


Figure 1. The standard figure format: native TikZ, shared colors and node styles, a compact title-free composition, and a caption that states the message.

1.2 State assumptions and diagnose them

Do not treat a model assumption as an isolated mathematical sentence. Trace it through the system that creates the model's inputs and evaluates its outputs.

Use emphasis selectively

Reserve colored boxes for a limitation, warning, or worked result that students should genuinely notice. Keep ordinary principles in ordinary prose.

When several layers must be compared exactly, use a LaTeX table:

Layer	Assumption	Evidence	Possible response
Problem	The target represents the real decision	Stakeholder definition and downstream outcome	Reframe the target
Representation	Available features preserve relevant information	Data audit and counterexamples	Revise collection or features
Model	The method's structural assumption is useful	Held-out behavior and error analysis	Change the ruler or method
Evaluation	The test reproduces future use	Prospective or deployment-aligned evidence	Redesign the validation

Table 1. The standard table format: native LaTeX, restrained rules, semantic columns, and a concise interpretive caption.

Lecture summary. Replace this with one sentence that connects the practical problem, the model's main assumption, and the evidence used to diagnose success or failure.

Further reading.

- ▶ the primary paper or canonical reference;
- ▶ official technical documentation; and
- ▶ the corresponding course notebook or project.

Acknowledgment. These notes are based on the instructor's handwritten notes and transcripts of lecture discussions and were editorially polished and typeset with assistance from OpenAI Codex and Anthropic Claude. The instructor reviewed and is responsible for the final content.